Database & Management System

[LAB MANUAL]

K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B.TECH CSE (CORE)

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LAB SHEET-3

Objectives

- To understand the use of COMMIT and ROLLBACK commands in transaction control.
- To use ORDER BY and GROUP BY clauses in SQL queries.
- To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.
- To understand and implement JOIN operations in SQL.

Instructions

1. Open your preferred Database Management System (DBMS) tool.
2. Create the required tables: TEACHER, SALES, DEPARTMENTS, and EMPLOYEES.
3. Insert suitable records into each table.
4. Execute the SQL commands for transaction control, privilege control, grouping, sorting, and joins.
5. Observe the output after each operation such as ROLLBACK, COMMIT, SAVEPOINT, GRANT, and REVOKE.

DDL Questions (Data Definition Language)

1. Create the **TEACHER** table with the following attributes:
   (Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).
2. Create the **SALES** table with the following attributes:
   (Sales_No, Salesname, Branch, Salesamount, DOB).
3. Create the **DEPARTMENTS** table with the following attributes:
   (dept_no, dept_name, dept_location).
4. Create the **EMPLOYEES** table with the following attributes:
   (emp_id, emp_name, emp_salary, dept_no).

DML Questions (Data Manipulation Language)

From TEACHER Table

1. Insert five records into the **TEACHER** table.
2. Give an increment of 25% salary to teachers working in the Mathematics department.
3. Give an increment of 15% salary to teachers working in the Commerce department.

From SALES Table

1. Insert five records into the **SALES** table.
2. Calculate the total sales amount in each branch.
3. Calculate the average sales amount in each branch.
4. Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.
5. Display the Salesname and DOB of salesmen in alphabetical order of the month.

From JOIN Operations

1. Display employee details and department details where the department numbers are the same in both tables.
2. Display the employee name and department name using a LEFT OUTER JOIN.
3. Display the employee name and department name using a RIGHT OUTER JOIN.
4. Display the details of employees who draw a salary greater than the average salary.

TCL Questions (Transaction Control Language

1. Perform the **ROLLBACK** command on the TEACHER table.
2. Perform the **COMMIT** command on the TEACHER table.
3. Write a query to implement a **SAVEPOINT**.

DCL Questions (Data Control Language)

1. Write a query to **grant all privileges** on the EMPLOYEES table to DEPARTMENTS.
2. Write a query to **grant selected privileges** on the EMPLOYEES table to DEPARTMENTS.
3. Write a query to **revoke all privileges** on the EMPLOYEES table from DEPARTMENTS.
4. Write a query to **revoke selected privileges** on the EMPLOYEES table from DEPARTMENTS.

Create Schema-

CREATE DATABASE experiment5;

USE experiment5;

48	00:01:24	CREATE DATABASE experiment5	1 row(s) affected	0.125 sec
49	00:01:24	USE experiment5	0 row(s) affected	0.000 sec

Create Tables-

TEACHER Table-

```
CREATE TABLE TEACHER (
    Name VARCHAR(50),
    DeptNo INT,
    Date_of_Joining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary INT
```

);

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Amit	10	2015-06-01	Mathematics	Delhi	40000
Neha	20	2016-07-15	Commerce	Mumbai	35000
Ravi	10	2017-08-20	Mathematics	Delhi	42000
Priya	30	2018-09-10	Science	Chennai	38000
Arjun	20	2019-10-05	Commerce	Pune	36000

teacher 1

<u>SALES Table-</u>

```
CREATE TABLE SALES (
    Sales_No INT PRIMARY KEY,
    Salesname VARCHAR(50),
    Branch VARCHAR(50),
    Salesamount INT,
    DOB DATE
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Sales_No	Salesname	Branch	Salesamount	DOB
1	Rahul	Delhi	5000	1990-12-21
2	Meera	Mumbai	7000	1988-11-15
3	Sohan	Delhi	6000	1992-12-05
4	Priya	Chennai	8000	1991-07-10
5	Vikas	Mumbai	5500	1993-01-25
NULL	NULL	NULL	NULL	NULL

sales 1

<u>DEPARTMENTS Table-</u>

```
CREATE TABLE DEPARTMENTS (
    dept_no INT PRIMARY KEY,
```

```
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

dept_no	dept_name	dept_location
10	Mathematics	Delhi
20	Commerce	Mumbai
30	Science	Chennai
NULL	NULL	NULL

departments 1

<u>EMPLOYEES Table-</u>

```
CREATE TABLE EMPLOYEES (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    emp_salary INT,
    dept_no INT,
    FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	emp_salary	dept_no
101	Amit	5000	10
102	Neha	4800	20
103	Ravi	6000	10
104	Priya	4500	30
105	Arjun	7000	20
NULL	NULL	NULL	NULL

employees 1

Insert Records-

INSERT INTO TEACHER VALUES

('Amit', 10, '2015-06-01', 'Mathematics', 'Delhi', 40000),

('Neha', 20, '2016-07-15', 'Commerce', 'Mumbai', 35000),

('Ravi', 10, '2017-08-20', 'Mathematics', 'Delhi', 42000),

('Priya', 30, '2018-09-10', 'Science', 'Chennai', 38000),

('Arjun', 20, '2019-10-05', 'Commerce', 'Pune', 36000);

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Amit	10	2015-06-01	Mathematics	Delhi	40000
Neha	20	2016-07-15	Commerce	Mumbai	35000
Ravi	10	2017-08-20	Mathematics	Delhi	42000
Priya	30	2018-09-10	Science	Chennai	38000
Arjun	20	2019-10-05	Commerce	Pune	36000

teacher 1

INSERT INTO SALES VALUES

(1, 'Rahul', 'Delhi', 5000, '1990-12-21'),

(2, 'Meera', 'Mumbai', 7000, '1988-11-15'),

(3, 'Sohan', 'Delhi', 6000, '1992-12-05'),

(4, 'Priya', 'Chennai', 8000, '1991-07-10'),

(5, 'Vikas', 'Mumbai', 5500, '1993-01-25');

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Sales_No	Salesname	Branch	Salesamount	DOB
1	Rahul	Delhi	5000	1990-12-21
2	Meera	Mumbai	7000	1988-11-15
3	Sohan	Delhi	6000	1992-12-05
4	Priya	Chennai	8000	1991-07-10
5	Vikas	Mumbai	5500	1993-01-25
NULL	NULL	NULL	NULL	NULL

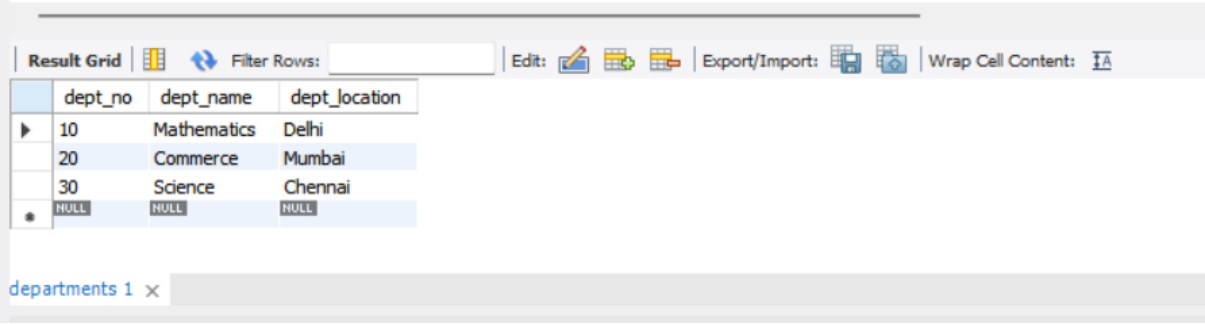
sales 1

INSERT INTO DEPARTMENTS VALUES

(10, 'Mathematics', 'Delhi'),

(20, 'Commerce', 'Mumbai'),

(30, 'Science', 'Chennai');



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

dept_no	dept_name	dept_location
10	Mathematics	Delhi
20	Commerce	Mumbai
30	Science	Chennai
NULL	NULL	NULL

departments 1

INSERT INTO EMPLOYEES VALUES

(101, 'Amit', 5000, 10),

(102, 'Neha', 4800, 20),

(103, 'Ravi', 6000, 10),

(104, 'Priya', 4500, 30),

(105, 'Arjun', 7000, 20);

DML Operations-

TEACHER Salary Updates-

UPDATE TEACHER

SET Salary = Salary * 1.25

WHERE DeptNo = 10;

UPDATE TEACHER

SET Salary = Salary * 1.15

WHERE DeptNo = 20;

SALES Queries-

Total sales amount per branch-

SELECT Branch, SUM(Salesamount) AS TotalSales FROM SALES GROUP BY Branch;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Branch	TotalSales
Delhi	11000
Mumbai	12500
Chennai	8000

Result 2

<u>Average sales amount per branch-</u>

SELECT Branch, AVG(Salesamount) AS AvgSales FROM SALES GROUP BY Branch;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Branch	AvgSales
Delhi	5500.0000
Mumbai	6250.0000
Chennai	8000.0000

Result 3

<u>Salesmen born in December (formatted date)-</u>

SELECT Salesname, DATE_FORMAT(DOB, '%d-%b-%y') AS DOB

FROM SALES

WHERE MONTH(DOB) = 12;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Salesname	DOB
Rahul	21-Dec-90
Sohan	05-Dec-92

Salesmen ordered by month of birth-

SELECT Salesname, DATE_FORMAT(DOB, '%d-%b-%y') AS DOB

FROM SALES

ORDER BY MONTH(DOB);

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Salesname	DOB
Vikas	25-Jan-93
Priya	10-Jul-91
Meera	15-Nov-88
Rahul	21-Dec-90
Sohan	05-Dec-92

JOIN Operations-

Employee details with department details (INNER JOIN)-

SELECT e.emp_id, e.emp_name, e.emp_salary, d.dept_name, d.dept_location

FROM EMPLOYEES e

JOIN DEPARTMENTS d ON e.dept_no = d.dept_no;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	emp_salary	dept_name	dept_location
101	Amit	5000	Mathematics	Delhi
103	Ravi	6000	Mathematics	Delhi
102	Neha	4800	Commerce	Mumbai
105	Arjun	7000	Commerce	Mumbai
104	Priya	4500	Science	Chennai

Result 6

LEFT OUTER JOIN-

SELECT e.emp_name, d.dept_name

FROM EMPLOYEES e

LEFT JOIN DEPARTMENTS d ON e.dept_no = d.dept_no;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name
Amit	Mathematics
Neha	Commerce
Ravi	Mathematics
Priya	Science
Arjun	Commerce

Result 7

<u>RIGHT OUTER JOIN-</u>

SELECT e.emp_name, d.dept_name

FROM EMPLOYEES e

RIGHT JOIN DEPARTMENTS d ON e.dept_no = d.dept_no;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name
Amit	Mathematics
Ravi	Mathematics
Neha	Commerce
Arjun	Commerce
Priya	Science

Result 8

<u>Employees with salary greater than average-</u>

SELECT emp_name, emp_salary

FROM EMPLOYEES

WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);

Transaction Control (TCL)-

SET autocommit = 0;

START TRANSACTION;

Example transaction-

INSERT INTO TEACHER VALUES ('Chris', 40, '2020-01-01', 'Arts', 'Delhi', 30000);

SAVEPOINT A;

ROLLBACK TO A;

COMMIT;

Data Control Language (DCL)-

Grant all privileges on EMPLOYEES to DEPARTMENTS-

GRANT ALL PRIVILEGES ON EMPLOYEES TO 'departments'@'localhost';

Grant all privileges on EMPLOYEES to DEPARTMENTS

GRANT ALL PRIVILEGES ON EMPLOYEES TO 'departments'@'localhost';

Grant selected privileges (SELECT, UPDATE)

GRANT SELECT, UPDATE ON EMPLOYEES TO 'departments'@'localhost';

Revoke all privileges

REVOKE ALL PRIVILEGES ON EMPLOYEES FROM 'departments'@'localhost';

Revoke selected privileges

REVOKE SELECT, UPDATE ON EMPLOYEES FROM 'departments'@'localhost';